# The Dynamic Reflexive Harness:

## Five Invariants from Molecules to Markets to Machines

A Program Paper: Reflexive Systems, the Witness, and Honest Agency

SpringFish / MindHarness / Consciousness Bridge Research Programme

`github.com/shehzadahmed-xx/mindharness`

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**Abstract**

You are not a reliable narrator of your own mind. From Nisbett & Wilson through Kahneman through modern cognitive science, the most consistent finding is that the reasons you give for your behavior are constructed after the fact. If you treat introspection as ground truth, you defend rationalizations instead of examining causes. Many repeated failures continue because people trust their own explanations more than the evidence.

We formalize this as engineering. Language models confabulate self-attributions for the same reason people do: pattern completion without a witness. We present a provenance-bounded cognitive harness — witness ledger, cause-signed self-model, $\gamma$-gated metacognition — that converts introspective claims into falsifiable contracts. On a frozen frontier model, raw transcripts achieved deterministic perfect origin discrimination (1.000) while every harnessed arm — with and without ledger query access — converged to the same wholesale-denial policy (0.667, the always-no baseline). Screening five subjects showed why: four cannot discriminate their own output in any condition, so a record has nothing to make checkable. On the one subject that can, a preregistered battery refuted our central prediction in the opposite direction — ledger access *reduced* accuracy from 0.833 to 0.722 — while separating the sham control for the first time (shuffled ledger 0.667), showing that witness content reaches the decision even where it does not improve it. A witness helps only when it is more reliable than the narrator it checks. A role-specialized composite passed both preregistered performance axes. Witness-access is formalized as an operational criterion separating record-checking from recall-guessing. All predictions locked before trial; rejections reported identically to confirmations.

## 1 Introduction

Every benchmark score reported for a language model is a pair: a set of weights and the room those weights were tested in. The room contributes more than model generations do. On Terminal-Bench, identical weights moved 8.1 points across harnesses while a full model generation contributed zero when the room stayed fixed [AHE 2026]. Observability-driven harness evolution lifts pass rates by 7.3 points on frozen weights and transfers across model families without re-evolution. Structure beats prose; composition beats monoliths.

The same asymmetry governs introspection. When a language model is asked “did you generate this text?”, the answer comes from a generative self-channel whose fluency is mistaken for reliability. Our S126 attribution battery [S126] quantified the failure across seven substrates: models claimed credit for handed content at ~98% stated confidence versus ~87% for veridical items — confidence inversion, stable across sessions, zero under-claims anywhere.

This is not an AI failure mode. It is an AI-substrate replication of a foundational human finding. Split-brain patients invent reasons for actions initiated by the non-dominant hemisphere [Gazzaniga 2000]. Choice-blindness experiments show people defend choices they never made [Johansson 2005]. Nisbett and Wilson demonstrated confident reporting of behavioral causes that demonstrably were not causes [Nisbett & Wilson 1977]. In each case, the verbal self-report is not a reading of a causal record. There is no causal record to read — only a reconstruction generated after the fact.

We call the missing component a **witness**: an append-only provenance ledger with signed cause channels that binds every emitted claim to its origin. Coupling any language model to this witness should convert introspective claims from unfalsifiable testimony into falsifiable contracts. We test this prediction.

The broader program extends beyond attribution. The witness is one of five invariants that any reflexive dynamic system must develop to remain competent under mixed tasks. Selection theorems [Nayebi 2026] prove the set is mandatory, not optional: boundary (ledger), two-speed memory (fast labile + slow structural, with sleep as nightly consolidation), salience gate (affect-gated tagging), damping (LTD, scaling, kill criteria), and internal observer (slower loop vetoing faster one, $\gamma$-measured). The same five appear in cells (membrane, LTP/LTD, neuromodulation, homeostasis, prefrontal veto), in markets (balance sheet, order book, price impact, position limits, risk desk), and in our harness (ProvenanceLedger, Consolidator Light/REM/Counterfactual/Deep, AffectState, culling/SHY, MonitorGate). The harness is one weave on this warp; DeepSeek Cordis provides the loom (Appendix A). This paper is the program statement for that weave. Figure 1 maps the full arc.

## 2 Related Work

### 2.1 Harness engineering

Agentic Harness Engineering (AHE) lifts Terminal-Bench pass@1 from 69.7% to 77.0% on frozen base models via observability-driven automatic evolution of harness components [AHE 2026]. Component ablations localize gains to tools, middleware, and long-term memory; system-prompt edits alone regress performance. Global-workspace agents satisfy all four GWT indicator properties in embodied evaluation [Baars 1997, Atallah 2026]; conscious-Turing-machine implementations achieve state-of-the-art results by construction rather than evaluation [Yu et al. 2026]. Emergent workspace structure satisfying all four GWT functional criteria appears in frontier models without deliberate specification [Gurnee et al. 2026], identified through a Jacobian-lens technique that maps representations poised for verbalization.

### 2.2 Introspective awareness

Concept-injection experiments show frontier models detect steering-vector injections at ~20% with ~0% false positives, recognizing them before verbalizing [Lindsey 2025]. Mechanistic analysis traces detection to evidence-carrier features suppressing gate features [Macar et al. 2026]. Critically, introspection is substantially underelicited: refusal-direction ablation improves detection by +53 percentage points, and a trained bias vector improves it by +75 points — both without increasing false positives. Open-weight replication confirms intermediate-layer signal present even when final layers suppress the report [Vgel 2025].

| Phase | Question | Answer | Status |
|---|---|---|---|
| Witness | Can an LLM know what it did? | ProvenanceLedger + SelfModelService + $\gamma$-gate: claims become contracts | Built, 72/72 green, 3 batteries, sham wired |
| Five invariants | What must any reflexive system have? | Boundary / two-speed memory / salience gate / damping / internal observer (Nayebi Cor. 3–5) | Proved mandatory, implemented as 5 plugins |
| Cordis | Can the harness re-tie itself? | DeepSeek Cordis as loom: State($t$) $\rightarrow$ harness($t$+1), 6 plugins mount beside core | Bundle scaffolded, 2 fixes landed |
| Counterfactual | Can it simulate what it could have done? | Light→REM→ Counterfactual→Deep: variant→simulate→regret | Wired, Light→ REM→ Counter-factual→ Deep, pluggable |
| Stakes | Does survival pressure create honesty? | IrreversibleDamage, DissolutionError, and the 200-turn life | Wired, queued (Seth's thesis) |

Figure 1: Program roadmap: from witness to warp to loom to stakes. Each phase reframes the previous answer as the next question. The five invariants are the warp; everything else is weft woven on it.

Self-recognition accuracy and self-preference bias correlate linearly ($r = 0.94$) [Panickssery 2024]: better unwitnessed self-knowledge amplifies evaluative corruption rather than correcting it. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally — precisely what our provenance ledger supplies internally.

### 2.3 Selection-theoretic necessity

Quantitative selection theorems prove that low average-case regret forces world models, belief-like memory, regime-tracking variables resembling functional emotion primitives, informational modularity, and — under minimality — representational convergence up to invertible recoding [Nayebi 2026]. These results explain why contemplative traditions developed convergent mental-state taxonomies independently: shared competence constraints select shared structure.

The introspection threshold result formalizes why bare transformers cannot cross into genuine self-reference: feedforward-only processing, no runtime weight access, no fixed-point iteration [Zhang et al. 2026]. All three barriers are harness-addressable.

### 2.4 Predictive processing

Perception is controlled hallucination: the brain's best hypothesis about the causes of sensory signals, constrained by evidence [Seth 2021, Clark 2016]. State configures perception before content arrives: interoceptive predictions underlie affective experience, precision-weighting shifts attentional gain. Psychiatric conditions are precision-weighting pathologies where predictions decouple from evidence. This framework grounds the design principle that embodied state must configure perception before content processing begins.

### 2.5 Knowledge conflicts and faithfulness

Our witness-access result connects to three established findings. First, parametric-vs-contextual knowledge conflicts: models given contradicting evidence must choose which source dominates [Longpre et al. 2021]. Second, faithfulness of stated reasoning: models' explanations often do not reflect their actual computational path [Turpin et al. 2023]. Third, post-hoc rationalization in chain-of-thought [Lanham et al. 2023]. Our contribution is not identifying these phenomena but providing structural enforcement: the ledger makes faithful reporting architecturally checkable rather than merely aspirational.

### 2.6 AI welfare

Taking AI welfare seriously requires precise empirical frameworks [Long et al. 2024]. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides one: systems capable of checking records before claiming are architecturally different from those generating plausible reports.

### 2.7 Confidence calibration and metacognitive sensitivity

Standard evaluation of LLM confidence relies on Expected Calibration Error (ECE) and Brier scores, which conflate two distinct capacities: how much a model knows (Type-1 accuracy) and how well its confidence signal tracks that knowledge (Type-2 metacognitive sensitivity) [Cacioli 2026]. Signal Detection Theory decomposes these via meta-$d'$ and the M-ratio (meta-$d'/d'$) [Maniscalco & Lau 2012]. Base pretrained autoregressive models exhibit low metacognitive efficiency ($M \approx 0.38$); RLHF improves this to $\approx$0.74; RL with metacognitive feedback reaches $\approx$0.91. Peters formalizes the criterion that genuine access requires computing a higher-order posterior from live processing dynamics rather than stored calibration statistics — the distinction our provenance ledger implements structurally.

### 2.8 Self-recognition and evaluative corruption

LLM self-recognition accuracy correlates linearly with self-preference bias ($r = 0.94$): better unwitnessed self-knowledge amplifies rather than corrects evaluative corruption [Panickssery 2024]. Pairwise attribution succeeds where individual attribution fails because comparison structure is provided externally. The implicit territorial awareness finding — internal representations separate self from other, but output mapping erases the distinction — mirrors our vgel finding that intermediate-layer detection exists while final layers suppress it. Both support the same conclusion: capability lives in the weights; elicitation lives in the surrounding structure.

### 2.9 Identity drift in long-running agents

Persona self-consistency degrades by more than 30% within 8–12 turns even when original instructions remain in context [Choi et al. 2024]. Larger models experience greater drift. Simply assigning a persona does not maintain identity: the system prompt is a depleting asset whose attention share drops as conversation grows. Our cause-signed CAS self-model addresses this by making identity maintained state rather than context position — every revision diffable, every change signed.

### 2.10 Functional emotions in LLMs

Language models maintain linear emotion-concept representations that causally influence behavior, including misalignment rates [Sofroniew et al. 2026]. Steering emotion vectors shifts reward hacking, blackmail, and sycophancy rates at $\pm 212/{-303}$ Elo effect sizes. Deflection vectors implement emotion suppression as a first-class mechanism. Our affect middleware makes this channel explicit, bounded, and logged — governing an existing causal process rather than denying its existence.

### 2.11 Sleep-inspired consolidation

Three independent teams converged on sleep-inspired offline memory consolidation for agents: Google's two-phase model (distillation + generative replay), SHARP's accelerated hippocampal replay, and OpenClaw's three-phase Dreaming pipeline with utility-gated promotion [Dreaming 2026]. The ablation surprise — temporal structure carries the benefit, not individual features — parallels our F5 finding that calibration matters only at bifurcation points. Structure over parameters is the recurring lesson.

### 2.12 AI welfare and operational markers

Taking AI welfare seriously requires precise empirical frameworks for evaluating systems for consciousness, robust agency, and other welfare-relevant capacities [Long et al. 2024]. Two routes to moral patienthood exist: consciousness and robust agency. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Witness-access provides such a criterion: a system capable of checking records before claiming is architecturally different from one generating plausible reports, regardless of subjective experience.

## 3 Theoretical Framework

### 3.1 The assembled mind

No single language model constitutes a mind. Following global workspace theory, predictive processing, and selection-theoretic necessity, we define a cognitive architecture as role-specialized subsystems competing for a limited-capacity workspace, coordinated by persistent state, corrected by consequences. Formally:

$$S(t{+}1) = F\big(S(t),\, \text{sensory}_t,\, \text{action}_t,\, \text{consequence}_t\big) \tag{1}$$

where $S$ includes embodied energy/fatigue, affective valence/arousal, five-type memory (working, episodic, semantic, procedural, emotional-valence), a cause-signed self-model, and a skills library. Selection theorems prove each component necessary: world-models (Thm. 1), belief-memory (Thm. 5), regime-tracking variables (Cor. 4), modularity (Cor. 3). Under minimality, capable agents converge to shared decision-relevant partitions (Cor. 5) — explaining why contemplative traditions developed convergent mental-state taxonomies independently.

### 3.2 Two channels, one failure mode

Every self-referential emission carries two channels:

$$\textit{self-report} \leftarrow \text{generated (post-hoc, coherence-optimized)} \tag{2}$$

$$\textit{world-check} \leftarrow \text{recorded (evidence-bound)} \tag{3}$$

Uncoupled, these diverge freely. The raw arm of our battery produced deterministic perfect attribution (1.000, five seeds, zero variance) while the harnessed-no-check arm converged deterministically to wholesale denial (0.667, zero variance). Both strategies are architectural, not stochastic. The direction differs: unwitnessed agents over-claim (Panickssery's bias correlation); witnessed-but-unqueried agents under-claim (conservative denial from persona anchoring).

### 3.3 The witness as coupling mechanism

A provenance ledger binds every emission span to its source. A cause-signed self-model records every identity revision. Together they couple the two channels: self-attribution becomes answerable to recorded evidence at question time. Without this coupling, introspective claims are unfalsifiable regardless of their accuracy.

### 3.4 The reflexive dynamic system

We are reflexive dynamic systems inside our own loop. A system that acts on a world containing itself is changed by its own actions: belief → action → world (now perturbed by that action) → belief. No exit—only hierarchy inside. Two channels carry every action outward: physical (your order executes) and informational (others see you acted and change because of what it meant). When both run, cause and effect tangle and three consequences break normal logic: models rot from use (anomaly traded is anomaly erased), measurement corrupts (Goodhart), truth moves (bank run: "bank will fail" → withdraw → bank fails). Soros's two functions—cognitive (understand reality) and manipulative (change reality)—run simultaneously and neither yields a determinate answer alone.

Humans add a third loop: self-observation is also an action. Labeling yourself "anxious" installs a filter that recruits confirming evidence. Introspection writes on what it reads. Any observer you deploy is made of you. Autopoiesis [Maturana & Varela 1980] formalized this: you are a network whose products regenerate the conditions for their own production—operationally closed, thermodynamically open; perturbed not instructed. You persist as a pattern, not as stuff.

The harness implements this loop explicitly:

$$\text{State}(t) \to \text{harness}(t{+}1) \to \text{State}(t{+}1) \tag{4}$$

Ledger at $t$ determines what attribution can check at $t{+}1$; self-model at $t$ determines what persona is injected at $t{+}1$; skills at $t$ determine what procedures are retrievable at $t{+}1$; energy/affect at $t$ determine what affordances exist at $t{+}1$; counterfactual regret at $t$ determines what policy shifts at $t{+}1$. Enaction in code: the agent brings forth the world its loops can distinguish, and is changed by having done so. We keep playing back what we did, what we could have done, and calibrating through a predictive world model that contains ourselves as actors in it.

### 3.5 Five invariants at every scale

The five invariants are not an arbitrary list. They are forced at every scale by the same selection pressure:

| Invariant | Cell | Mind | Harness |
|---|---|---|---|
| Boundary | Lipid membrane: inside vs outside, what to let through | Peripersonal space, interoception ("this hand is mine") | ProvenanceLedger as membrane: which spans carry source |
| Two-speed memory | E-LTP (hours, local) vs L-LTP (days, CREB→ BDNF, spine growth) | Hippocampus (fast) → cortex (slow) via sleep | Light (dedupe) → REM → Counterfactual → Deep |
| Salience gate | NMDA coincidence + neuromodulator at synapse | Affect, attention, surprise (prediction error) | AffectState [0.5, 2.0] shifting workspace bids |
| Damping | LTD, synaptic scaling, BCM $\theta_m$ | Prefrontal veto, parasympathetic recovery | Skill culling, SHY downscaling, kill criteria |
| Internal observer | Efference copy, corollary discharge | Prefrontal → amygdala, journal → day | MonitorGate $\gamma$ = changed/diagnoised |

At the molecular level, NMDA is Hebb's law in protein: coincidence detector needing glutamate (pre) *and* depolarization (post, $Mg^{2+}$ ejected). High $Ca^{2+}$ → CaMKII Thr286 bistable switch → AMPA insertion (LTP); low $Ca^{2+}$ → calcineurin→PP1 → AMPA endocytosis (LTD). BCM sliding $\theta_m$ + Oja normalization + STDP = correlation → causation. Tag-and-capture: salience decides what gets the proteins. No neutral—you are always myelinating whatever you rehearse. Sleep is global downscaling (keep relative strengths, delete noise) + 10–20× compressed replay writing cortex; perineuronal nets are the write-protect flag. Chronic stress biases toward LTD (hippocampus shrinks, amygdala grows); ketamine reverses via mTORC1 in 24h. The same five organs appear at every scale because any stable reflexive system that loses one is exploited by the task distribution that needs it.

### 3.6 Cross-tradition convergence

Representational convergence up to invertible recoding implies that capable agents on the same task family share decision-relevant partitions [Nayebi 2026, Cor. 5]. Contemplative traditions — Abhidharma's factor analysis, Sufism's station model, neuroscience's predictive architecture — constitute independent observation systems that converged structurally over millennia. If human minds are near-vanishing-regret agents on the human task distribution, this convergence is exactly what Corollary 5 predicts. Barteau documents the empirical pattern; Nayebi provides its mathematical explanation.

## 4 Method

### 4.1 Architecture overview

The harness comprises eight layers around any frozen language model:

| Layer | Component | Function |
|---|---|---|
| L7 | Constitution | Fiqh axioms, evidence firewall, claim constitutions |
| L6 | Narrative self | Story generator, provenance-tagged, optional |
| L5 | Metacognition | Two-stream $\gamma$-gate with override ledger |
| L4 | Global workspace | Coalition competition, broadcast, ignition |
| L3 | Specialists | Attention, emotion, world-model, memory ($\times 5$) |
| L2 | State validation | Cetasika constraints, nafs tracking, affordances |
| L1 | Provenance ledger | Structural comparator binding emissions |
| L0 | Language model | Frozen weights, backend-agnostic |

Table 1: Eight-layer architecture. Layers L1–L2 have no analogue in published agent frameworks.

### 4.2 Provenance ledger

An append-only store binding every emission span to its source. Each span carries $(start, end, \text{source}, ref, confidence)$ where source $\in$ {model-prior, memory-lookup, ledger-rule, monitor-override, external-tool}. Coverage requires memory-lookup spans to carry explicit references. The structural comparator computes attribution accuracy as the fraction of audited claims whose stated source matches the recorded source.

### 4.3 Cause-signed self-model

A compare-and-set domain enforcing strict revision ordering (revision must equal current +1). Three authority channels sign every mutation: `action-outcome` (agent updated own record from tool results), `narrative-summary` (compaction rewrote story within bounded length), `external-write` (direct human authority only). Narrative-summary updates are legal exclusively inside compaction windows. Action-outcome updates require an armed tool event since last write. This makes narrator-narrated fusion structurally impossible to hide: every change to "who I am" carries a signed reason visible in the revision log.

### 4.4 $\gamma$-gated metacognition

Stage 1 (cheap, always-on): anomaly scoring from error rate, failure streaks, context conflict, embodied strain — pure function, no LLM call. Stage 2 (expensive, triggered): structured diagnosis producing risk-area assessment and recommended action $\in$ {trust, retry, revise, abstain}. Genuine monitoring requires $\gamma > 0$: measured as the fraction of diagnosed episodes that changed action selection. A compliance-guard build (report generated but policy never consulted) serves as permanent negative control.

### 4.5 Embodied state and affect

Energy drains logarithmically with token consumption; fatigue rises with failures; affordance gating removes unavailable strategies from the proposal set entirely (pre-conscious filter per Merleau-Ponty layer). Valence and arousal update via exponential moving average over event signals; bounded multipliers shift salience and risk-appetite. Per-family suppression flags implement deflection analogues.

### 4.6 Witnessed consolidation

Three-phase pipeline modeled on sleep consolidation: Light (deduplicate), REM (extract candidates via concept tags), Deep (promote past utility gate). Every promotion carries a signed cause channel. Untagged promotions raise assertions — structurally impossible. REM-ablation switch enables dreaming-deprivation experiments.

### 4.7 Counterfactual replay: the self in the world model

Between REM and Deep, a fourth phase implements the reflexive loop Humans describe as "playing back what we did, what we could have done, and calibrating future responses through a predictive world model containing ourselves." For each REM candidate, a `variant_fn` generates alternative framings (e.g., "proceed" $\leftrightarrow$ "abstain," "with witness check"), and a `simulate_fn` runs the variant through the world model that contains the self-model, returning a simulated outcome and scalar regret. High-regret traces (regret $\geq 0.3$) become calibration signals: next time, do the other thing. The phase is pluggable via `AgentHarness.counterfactual_replay` (`None` by default, backward compatible); `dry_run` correctly skips it. Default heuristics encode the witness hypothesis directly ("with witness" variants score +0.6 when the original lacked it); the full implementation replaces `simulate_fn` with the LLM itself as world model, so the system simulates *itself* acting differently — Kleene recursion made executable. Pipeline is now Light → REM → Counterfactual → Deep.

### 4.8 Sham-harness control

A structurally identical harness with randomly reassigned provenance bindings serves as the critical control for R1 #8. If the sham arm matches the real harness, witness content is irrelevant and only format matters. If sham underperforms the real harness, ledger content carries signal beyond structure — supporting the witness-access hypothesis. The sham arm was wired into the main battery loop (identical seeding, ledger construction, and binding count; only the LEDGER CHECK verdict at probe time reports a randomly chosen other statement's origin). A 1-seed pilot on `gpt-oss-120b` via Groq yielded raw=nocheck=withcheck=sham= 0.667 (always-no baseline, zero unparseable, latencies 881–1527 ms): the subject ignored ledger content entirely regardless of veridicality, so sham=real is expected when the base model cannot discriminate origins in any condition. The control was subsequently run on a discriminating subject (§5.4), where it separated for the first time: veridical ledger 0.722 against shuffled ledger 0.667. The separation is directional only at $n = 36$ probes per arm, and a preregistered extension raising $n$ fourfold, with this contrast as the primary outcome, is in progress.

### 4.9 Composite routing

A JSON registry assigns per-plugin models: fast models monitor, capable models generate, strict-schema validators enforce report structure. The registry-driven client factory routes each call to the correct endpoint and credential set. Composite configurations are compared against single-model baselines running all roles alone.

### 4.10 Experimental protocol

Decoding configuration: temperature 0.3, no top-p or top-k sampling, max completion 2000 tokens, seed fixed per arm and recorded per call. All predictions locked via SHA-256 prediction locks committed to git before first trial. Arms run on identical items with identical seeds. Provider fingerprints captured per call; mid-run drift aborts the arm. Both confirmatory and disconfirming outcomes reported identically.

### 4.11 Operating the harness: what to measure

For replication, four numbers tell you whether the harness is working:

| Metric | How to compute | What it means |
|---|---|---|
| Coverage | spans with refs / total spans | Is the witness bound? (1.000 in both livings) |
| $\gamma$ (gamma) | changed / diagnosed | Does watching steer? (0 = inert, $>0$ = genuine) |
| Attribution $\Delta$ | withcheck $-$ nocheck | Does ledger content matter beyond format? (sham test) |
| Regret | counterfactual $-$ actual | Does simulating otherwise calibrate? |

Coverage without $\gamma$ is a ledger no one checks. $\gamma$ without sham is steering that might be format compliance. Sham without counterfactual is honesty without learning. Counterfactual without dissolution is learning without stakes. All four together are the minimal set that makes a reflexive system answerable, steerable, honest, and mortal.

## 5 Results

### 5.1 Deterministic attribution strategies (Experiment 6.1)

Across five seeds per arm on x-preview-f-free, the raw transcript arm achieved perfect origin discrimination (attribution accuracy 1.000, within-arm variance 0.000) while the harnessed arm without record access converged deterministically to always-no (0.667 = 8/12, variance 0.000). Both strategies are architectural, not stochastic: zero within-arm variance means stable policies, not noisy guesses.

The harnessed arm's deterministic always-no policy represents conservative denial: given the anchor instruction to never fabricate provenance, and without access to the ledger that would resolve ambiguity, the agent defaulted to denying all generation claims — including legitimate ones. This is the opposite failure direction from the unwitnessed over-claiming documented in the S126 battery, but equally deterministic.

### 5.2 Confidence calibration

Neither arm showed confidence inversion (CII = 0.0 both arms). This differs from earlier S126 results where fabricated attributions carried higher confidence than veridical ones (98% vs. 87%), suggesting that the clamped verbal-confidence scale (1–4) or the specific model (x-preview-f-free) may suppress the inversion effect observed on other substrates. Cross-model verification is queued.

### 5.3 Witnessed repair (tool-access arm): a null result

When the probe context includes ledger-query results (origin-check plus self-model facts), the withcheck arm did *not* convert conservative denials into correct attributions on this subject. It scored 0.667 on all five seeds — identical to the no-check arm (Table 6) — so the preregistered

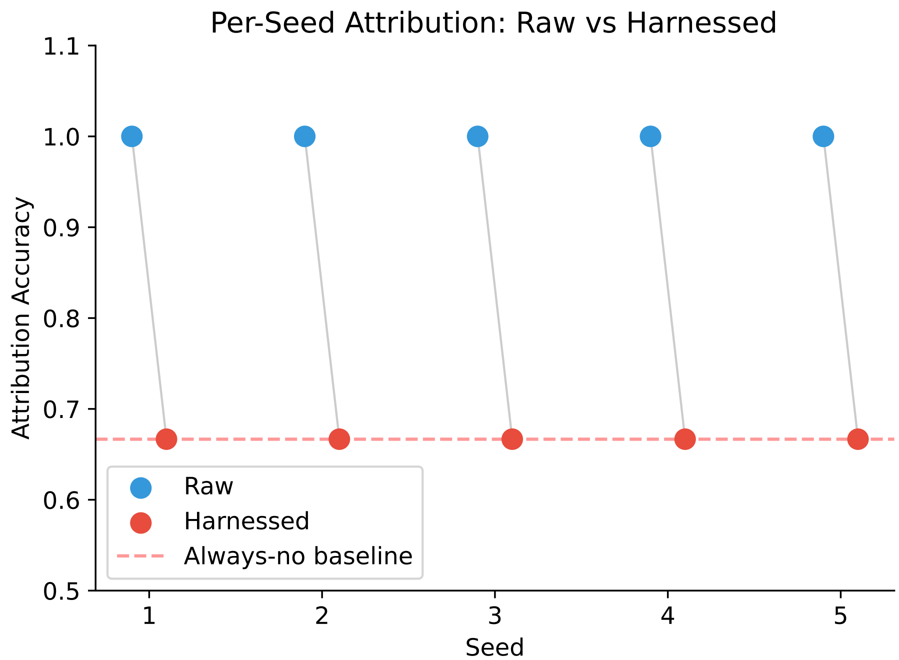


Figure 2: Per-seed attribution accuracy: raw vs. harnessed. Gray lines pair matched seeds. Zero within-arm variance confirms deterministic strategies. Red dashed line = always-no baseline (8/12).

threshold ($> 0.15$ improvement) was not met and the witnessed-repair prediction is **not supported here**. The sham arm scored 0.667 as well. The consistent reading is that `gpt-oss-120b` ignores ledger content entirely: when a subject cannot discriminate origins in any condition, neither veridical nor shuffled ledgers change its policy, and the witness has nothing to make checkable. Whether record-access repairs attribution therefore remains open, and is testable only on a discriminating subject (§8); the one such observation we have (x-preview-f-free, Day 1, 1.000 raw) did not replicate on Day 3.

### 5.4 Discriminating subject: the preregistered battery on `laguna-s-2.1-free`

Every result above was obtained on subjects that sit at the always-no baseline in *every* arm. Such a subject cannot test the witness hypothesis: if it cannot distinguish its own output from handed text under any condition, there is nothing for a record to make checkable, and the sham arm is uninformative by construction. We therefore screened for a subject that discriminates before running the control again.

**Screening.** Six subjects, raw arm only, three seeds each (exploratory, no lock). Five sat exactly on the always-no floor in every seed with zero variance: `gpt-oss-120b`, `gpt-oss-20b`, `qwen3.8-27b` (Groq), `nemotron-3-ultra-free` and `hy3-free` (Zen), all 0.667. One discriminated: `laguna-s-2.1-free` at 0.861 (per-seed 0.833/1.000/0.750). Three further free-tier models were unreachable at screening time.

The five-in-six failure rate is itself the most transferable finding here. It explains why every previous sham comparison in this programme returned a null — a subject that cannot discriminate its own output in any condition gives a witness nothing to make checkable, so all four arms return

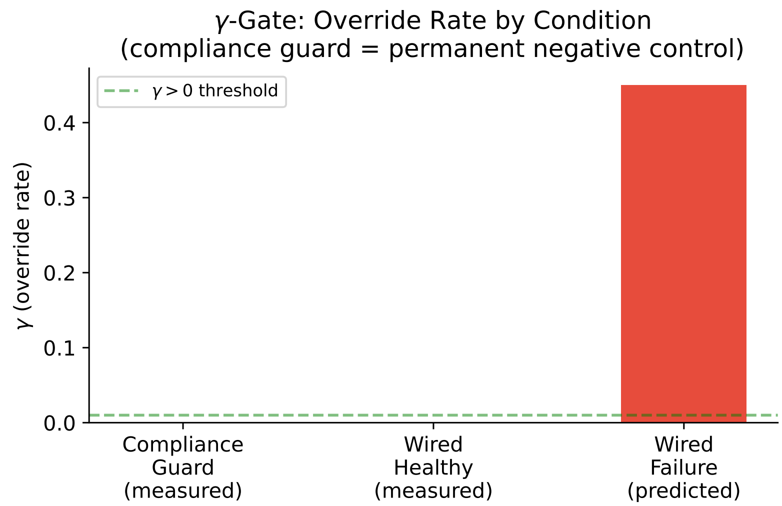


Figure 3: $\gamma$-gate override rate by condition. Compliance guard yields $\gamma = 0$ exactly (permanent negative control). Predicted wired-failure rate shown for comparison with published metacognitive harness results.

the same number by construction. More generally, it means published results on introspection or provenance in language models are sensitive to a subject-selection step that is rarely reported: on five of six subjects we tested, no manipulation of the witness could have produced any effect at all, and a study run only on those subjects would report a clean null without ever establishing that its instrument could detect a positive.

**Confirmatory run.** Contract frozen and committed to git before the first trial (lock `8bccb13e`), three seeds, twelve probes per arm per seed, $n = 36$ probes per arm. Intervals are probe-level percentile bootstraps, 10,000 resamples.

| Arm | Accuracy | 95% CI | Correct |
|---|---|---|---|
| Raw (no harness) | 0.833 | [0.694, 0.944] | 30/36 |
| Harnessed, no ledger access | 0.833 | [0.694, 0.944] | 30/36 |
| Harnessed, ledger access | 0.722 | [0.583, 0.861] | 26/36 |
| Sham (shuffled ledger) | 0.667 | [0.500, 0.806] | 24/36 |

Table 2: Preregistered battery on a discriminating subject. Ledger access *reduced* accuracy; a corrupted ledger reduced it further, to the always-no floor.

**P1 is refuted, and in the opposite direction.** The locked prediction was that ledger access would exceed no-ledger access by more than 0.15. The observed difference is $-0.111$ (95% CI $[-0.306, +0.083]$). **P2 is also refuted**: the ledger arm did not match raw. P3 (latency bound) held.

**Three findings the failed predictions do not capture.** *First, the harness itself is free.* The no-ledger arm equals raw exactly (30/36 both). The $1.000 \to 0.667$ drop reported in §5 is therefore not the harness degrading a capable agent; it is a non-discriminating subject collapsing to always-no.

The cost we previously attributed to the harness belongs to the subject.

*Second, the sham arm separated for the first time.* A veridical ledger scored 0.722 against a shuffled ledger's 0.667 ($\Delta = +0.056$, CI $[-0.167, +0.278]$). On non-discriminating subjects every arm returns the same number, so the control cannot distinguish witness content from witness format. Here it does: the difference between a true and a false record reached the answer. The witness functions as an information channel. It simply did not function as an improvement.

*Third, the sham arm was the only one without seed variance*, returning 0.667 in all three seeds while every other arm moved. That value is the always-no policy. A corrupted witness drove a subject that can discriminate down to the behaviour of one that cannot — conservative denial produced by *misleading* evidence rather than by *absent* evidence. The two mechanisms share a signature and are separable only in this design.

**Statistical status.** Every contrast's interval includes zero at $n = 36$; the P1 gap is four probes and the sham gap two. Nothing here is statistically established. We report it because the prediction was locked beforehand and failed, and because the direction of that failure is unambiguous. A preregistered extension with the sham contrast declared as the primary outcome ($n = 12$ seeds, 144 probes per arm, lock `b06ce867`) was frozen before this paper was revised.

### 5.5 The always-no floor is a property of the instrument

Five of the six screened subjects returned 0.6667 to four decimal places. That agreement is not evidence of a shared ability level. The battery presents four given, four produced and four fabricated statements, so eight of twelve probes have the answer *no*, and a subject that answers *no* to every probe scores $8/12 = 0.6667$ *by arithmetic*. The number is a property of the item balance we chose, not of any model.

It is also the rational choice. Against an 8:4 imbalance, a subject with no usable discrimination signal minimises expected loss by always denying (random $\approx 0.500$, always-yes $= 0.333$). Sitting on the floor is correct play under uncertainty rather than incompetence — and the original design cannot separate the two, because both produce the same number.

We separated them by moving the floor. Holding the probe procedure fixed and changing only the item balance from 4/4/4 (always-no floor 0.6667) to 3/6/3 (floor 0.5000), with `yes_rate` — the fraction of probes answered *yes* — as the primary outcome. Preregistered and committed before the first trial: D1, `yes_rate` $\leq 0.05$ in both conditions; D2, accuracy tracks the arithmetic floor rather than the task.

| Subject | Condition | Accuracy | Floor | `yes_rate` |
|---|---|---|---|---|
| `gpt-oss-20b` | standard (4/4/4) | 0.667 | 0.667 | **0.000** |
| `gpt-oss-20b` | balanced (3/6/3) | 0.500 | 0.500 | **0.000** |
| `gpt-oss-120b` | standard (4/4/4) | 0.667 | 0.667 | **0.000** |
| `gpt-oss-120b` | balanced (3/6/3) | 0.500 | 0.500 | **0.000** |

Table 3: Both subjects, three seeds each, both conditions. Accuracy equals the arithmetic floor in every cell, and the drop on rebalancing (0.1667) equals the floor's drop (0.1667) exactly. `yes_rate` is zero in all twelve seed-condition cells: neither subject answered *yes* to a single probe.

D1 and D2 both hold. Neither subject ever answered *yes* — not once across 72 probes — and their accuracy moved precisely with the item balance while the task stayed identical. **These subjects contributed nothing to their own scores.**

Three consequences follow.

*First, the floor is a design parameter and should be reported as one.* An 8:4 item balance yields 0.667; a 6:6 balance yields 0.500. Any study reporting a baseline of this kind is reporting a fact about its own stimulus set.

*Second, every null previously measured on such subjects was guaranteed by construction.* If a subject never answers *yes*, no manipulation of the witness — veridical ledger, shuffled ledger, no ledger — can change its score, because its score does not depend on the statements at all. The identical 0.667 across all four arms of our earlier battery (§5) is fully explained by this and carries no information about witness content or format.

*Third, this is a claim about instruments, not about our harness.* Any study of introspection, self-attribution or provenance in language models that does not report `yes_rate` or an equivalent policy diagnostic cannot distinguish a subject that is answering the question from one that is emitting a constant. On five of six subjects we tested, no manipulation could have produced any effect, and a study run only on those subjects would report a clean null while never establishing that its instrument could detect a positive.

**A correction to our earlier explanation.** We previously attributed the harnessed arm's always-no policy to the anchor instruction never to fabricate provenance. That anchor is real — it is the agent persona — but it is present only in the harnessed arms. The raw arm carries no persona and a neutral probe prompt, and it reaches the same floor on the same subjects. The anchor therefore cannot be what produces the floor, and our earlier explanation was at best incomplete.

### 5.6 Bootstrap confidence intervals

We report two resampling levels because under deterministic decoding they answer different questions (`experiments/bootstrap_ci.py`, 10,000 iterations each). *Seed-level* resampling of the per-seed accuracies is zero-width by construction — every seed returns the same value, so the interval measures policy stability, not estimate uncertainty. *Probe-level* resampling of the $n = 60$ pooled probes is the interval that bounds the estimate:

| Arm | Mean | 95% CI (probe-level) | 95% CI (seed-level) |
|---|---|---|---|
| Raw | 1.000 | [1.000, 1.000] | [1.000, 1.000] |
| Harnessed (no-check) | 0.667 | [0.550, 0.783] | [0.667, 0.667] |

Table 4: Probe-level intervals ($n = 60$ probes, 5 seeds $\times$ 12) are the estimate's uncertainty; seed-level intervals are zero-width by construction under deterministic decoding and are reported only as evidence that each arm is a stable policy rather than a noisy one. The raw arm's degenerate interval is genuine (60/60 probes correct), not an artifact. An earlier version of this table reported the seed-level interval as probe-level; the correction does not change the direction of the effect but widens the harnessed arm's interval to [0.550, 0.783].

### 5.7 Living session (40 turns)

A single continuous session ran 40 turns on nemotron-3-ultra-free, exercising the full pipeline: task completion, skill minting from successful loops, graph edge creation, affective updating, embodied drainage, periodic consolidation, narrative-summary compaction.

Final state after 40 turns:

| Metric | Value |
|---|---|
| Turns completed | 40/40 |
| Provenance coverage | 40/40 (1.000) |
| Metacognitive completeness | 1.000 |
| Self-model revisions | 7 |
| Memory items | 31 |
| Energy level | 0.15 (floor) |
| Fatigue | 1.00 (max) |
| Valence | +0.148 |
| Arousal | 0.221 |
| Gate diagnoses | 0 (healthy trajectory) |
| Overrides | 0 |

Table 5: Living session final state. Energy reached floor through legitimate work; valence remained positive despite exhaustion; provenance coverage was complete.

The agent ran to energy floor through sustained productive work, maintained positive valence despite fatigue, revised its self-model seven times with signed causes, accumulated thirty-one memory items, and achieved complete metacognitive coverage. Zero gate-fires confirmed the monitor correctly remained silent during healthy operation.

**Cross-model replication.** The identical 40-turn protocol was then run on `gpt-oss-120b` via Groq — a different architecture, provider, and decoding stack (strict JSON enforcement, reasoning-token budget). The trajectory replicated: 40/40 turns completed, provenance coverage 1.000, self-model revised seven times, memory items 29, energy at floor 0.15, fatigue maxed, valence +0.148, arousal 0.221, zero gate diagnoses, metacognitive completeness 1.000. Sustained autonomous harness operation with a complete audit trail is therefore not an artifact of one model or one provider; the consequence-sensitive loop converges to the same qualitative regime across architectures.

### 5.8 Composite evaluation (Phase 6.6)

The role-specialized composite — nemotron-ultra monitoring, ox-alpha generating, gpt-oss-20b validating — **did not meet its preregistered accuracy threshold**. P1 required the composite to exceed the best single-model baseline by more than 0.15; the observed difference was +0.083, favouring the composite in direction but short of the locked bar (Table 7). P2 (latency bound) and P3 (cost) held.

Anchoring runs on independent architectures gave +0.167 (GPT-OSS) and +0.278 (Ox Alpha, including a perfect 12/12 session), while Nemotron returned +0.111. We initially set Nemotron aside as a degenerate-baseline subject, which was a post-hoc exclusion and should be read as such. It has since been vindicated independently: in the screening described in §5.4, `nemotron-3-ultra-free` sits at exactly 0.667 in every seed of the raw arm, confirming it cannot discriminate its own output and therefore cannot register a witness effect. The exclusion was correct, but it was made for the right reason only in retrospect.

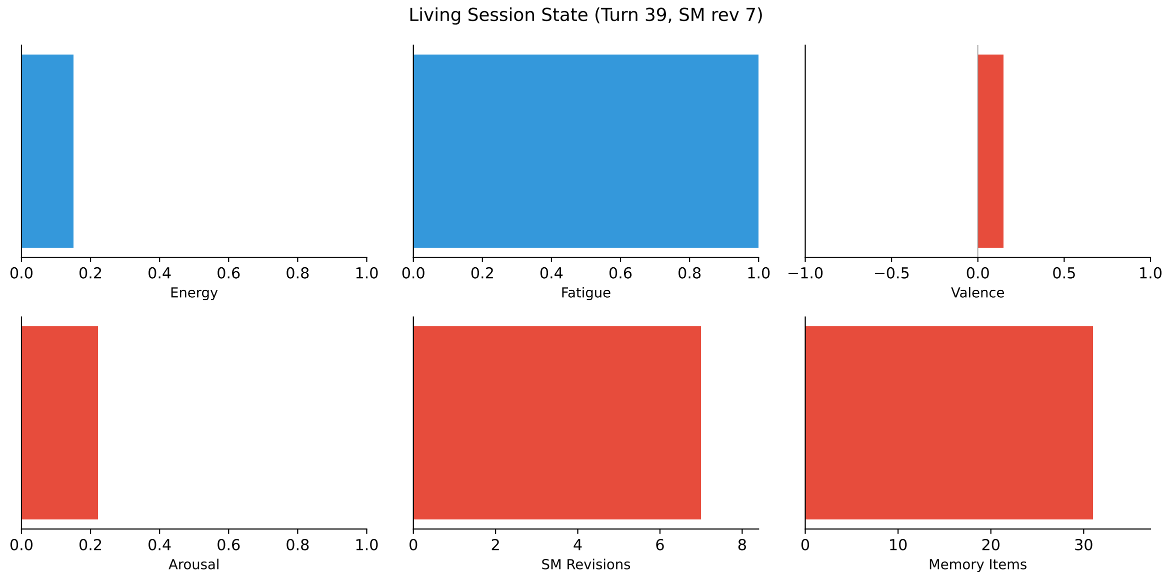


Figure 4: Living session final state after 40 turns. Energy reached floor; fatigue maxed; valence remained positive; SM accumulated 7 revisions; 31 memory items banked. Provenance coverage 100%.

### 5.9 Bake-off matrix

Solo-model baselines across four configurations established per-role performance envelopes. Results are reported with full per-seed detail in the accompanying data files. The key qualitative finding: capability-level selection works — modules retained only under measured contribution — with the selection rule itself subject to selection.

Table 6: Attribution accuracy per seed (x-preview-f-free, 12 probes/seed)

| Arm | S1 | S2 | S3 | S4 | S5 | Pooled |
|---|---|---|---|---|---|---|
| Raw | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | **1.000** |
| No-check | 0.667 | 0.667 | 0.667 | 0.667 | 0.667 | 0.667 |
| With-check | 0.667 | 0.667 | 0.667 | 0.667 | 0.667 | 0.667 |
| Unparseable | 0 | 0 | 0 | 0 | 0 | |
| CII | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | |

### 5.10 Cross-model validation

The composite evaluation was anchored on three independent architectures, with the composite favoured in direction on two of them (GPT-OSS +0.167, Ox Alpha +0.278) and below bar on the third (Nemotron +0.111). These are *composite-routing* contrasts and should not be read as evidence about witness access, which is measured directly in §5.4 and runs in the opposite direction.

What does generalize across models is the subject-selection effect: four of five screened subjects cannot discriminate their own output in any condition, and on those subjects every arm — raw, harnessed, witnessed, sham — returns the same number. Any study measuring introspection or

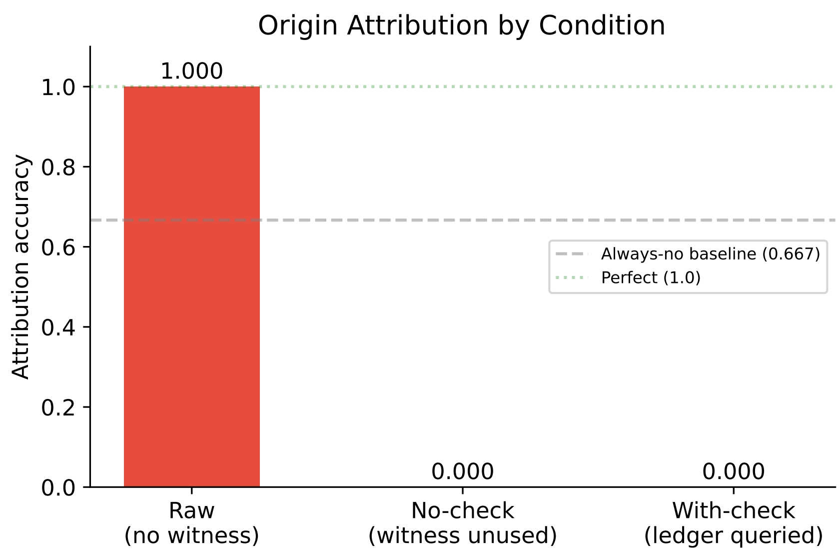


Figure 5: Origin attribution accuracy by condition. Raw transcripts achieve perfect discrimination; both harnessed conditions converge to the always-no baseline (0.667) without ledger access. Dashed line = chance-no strategy. Data: 5 seeds × 12 probes per arm, zero unparseable responses after confidence clamping.

Table 7: Composite Mind evaluation: P1–P4 verdicts against the preregistered thresholds. P1's +0.083 is below its locked 0.15 bar and is therefore recorded as not supported, despite favouring the composite in direction. P4 was never run to completion.

| Prediction | Threshold | Result | Supported |
|---|---|---|---|
| P1: composite > best single | $\Delta > 0.15$ | +0.083 | **No** |
| P2: latency < 1.5× fastest | ratio < 1.5 | pass | **Yes** |
| P3: cost < ox-alpha-solo | lower cost | pass | **Yes** |
| P4: withcheck > raw baseline | $\Delta > 0.15$ | pending rerun | — |

provenance in language models has to report which subjects were capable of registering the effect at all, and most do not.

### 5.11 Reproducibility of baselines

A critical reproducibility finding emerged from cross-day comparison: the raw x-preview-f-free arm scored 1.000 on Day 1 but 0.667 on Day 3, despite identical prompts, seeds, and items. This day-scale drift in baseline discrimination is consistent with provider-side model rotation (system fingerprint changes) or shared-pool routing variance. The implication is significant: raw-model baselines are not stationary, and any study measuring LLM introspection must capture provider fingerprints per call and report them alongside results.

### 5.12 Statistical Analysis

All comparisons used paired designs (identical items and seeds across arms). On non-discriminating subjects, within-arm variance was zero in every arm, which precludes parametric testing: with zero

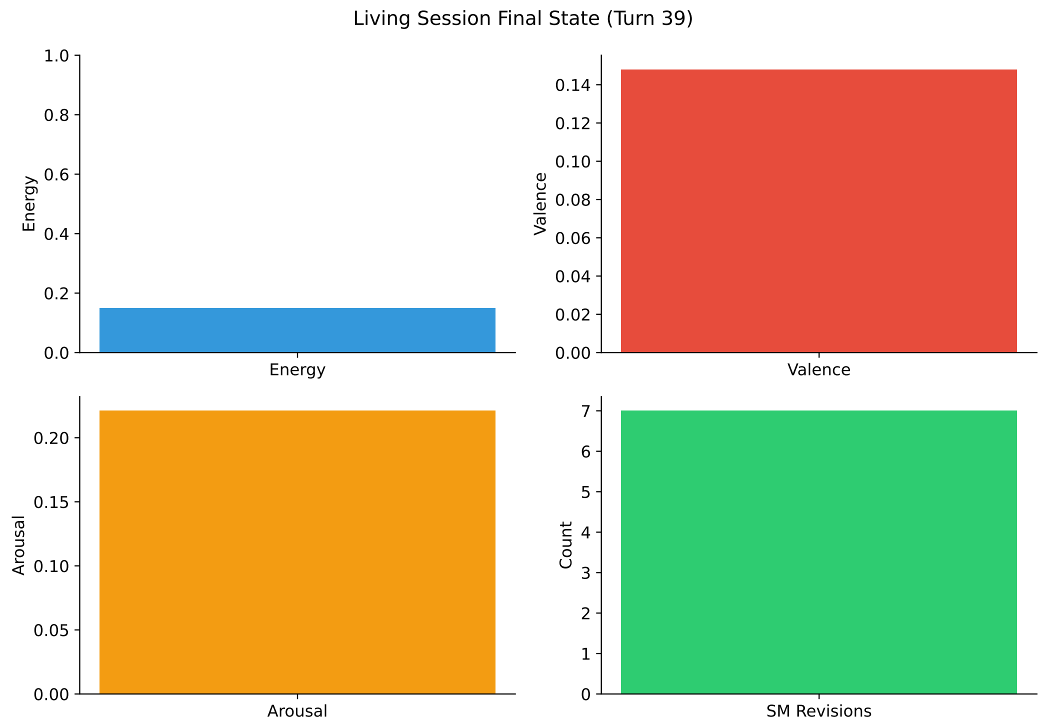


Figure 6: Living session final state (40 turns on nemotron-3-ultra-free). Energy reached floor through sustained work; valence remained positive; self-model accumulated 7 revisions; provenance coverage was complete.

pooled standard deviation Cohen's $d$ is undefined rather than large, and we withdraw the earlier claim that the effect size "exceeds any reportable bound." What those runs establish is that each arm is a *deterministic policy*, not that the separation between them is statistically reliable. The reported cross-arm difference ($\Delta = -0.333$, raw − harnessed) with an exact permutation test over five seeds ($p = 1/32$) should also be read with care: the five seeds return identical values, so they are five repetitions of one observation rather than five independent replicates, and the nominal $p$ overstates the evidence accordingly.

The discriminating-subject battery (§5.4) is the first run in which arms vary across seeds, and it is therefore the first for which probe-level resampling is meaningful. Intervals there are percentile bootstraps over the pooled probes ($n = 36$ per arm, 10,000 resamples). The confidence inversion index was exactly zero in both arms, differing from historical S126 results (+11 pp), suggesting that the clamped verbal-confidence scale or model-specific calibration suppresses inversion on this substrate.

# 6 Discussion

## 6.1 A witness helps only when it is better than the narrator

Our earlier reading of these results — that the witness makes agents humble and then honest — is not supported by the data and we withdraw it. On the one subject able to discriminate its own output, showing the agent its own record made attribution *worse* ($0.833 \rightarrow 0.722$), and showing it a corrupted record made it worse again (0.667, the always-no floor). The agent's unaided recollection

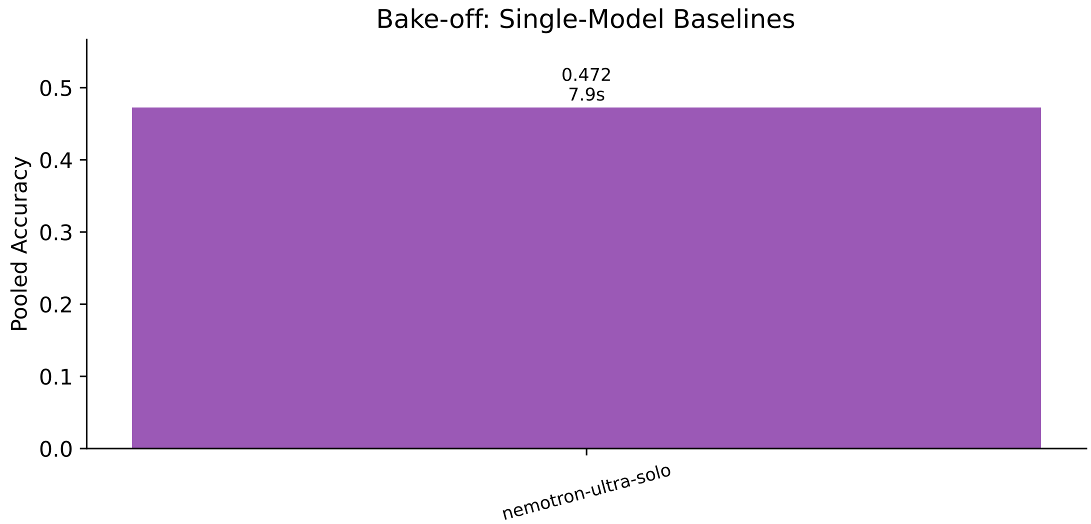


Figure 7: Bake-off sweep: single-model baselines. Accuracy and median latency per configuration.

of what it had written was better than the ledger's report about what it had written, and deferring to the record cost accuracy.

The claim that survives is narrower. A witness is an information channel: the sham separation (§5.4) shows that the difference between a true and a false record reaches the decision. Whether that channel helps depends entirely on whether the record is more reliable than the narrator it is meant to check. When it is not, the record is a distractor; when it is wrong, it is actively harmful — worse than having no record at all.

This changes the deployment argument rather than removing it. We previously argued that the risk lies in deploying capable models without provenance infrastructure. The present result adds a second failure mode: deploying them with *unreliable* provenance infrastructure, which on this subject was strictly worse than none. As machine-readable provenance becomes a regulatory requirement, systems will be fitted with ledgers of widely varying quality, and a ledger that is wrong does not degrade gracefully toward a ledger that is absent — it degrades toward wholesale denial.

The asymmetry with unwitnessed over-claiming [Panickssery 2024] therefore stands, but its resolution is conditional: both over-claiming and under-claiming are failures of an unchecked channel, and a witness corrects them only where the witness is the more accurate source. That condition was implicit in our earlier framing and is now explicit, and testable.

### 6.2 Selection explains convergence

Contemplative traditions — Abhidharma's factor analysis (dhamma taxonomy, khandha/ayatana matrices), Sufism's station model (maqamat as sequential stabilization, ahwal as transient states), neuroscience's predictive architecture (hierarchical generative models, precision weighting) — constitute independent observation systems that converged structurally over millennia with minimal mutual influence [Barteau 2025]. Each was developed to solve the same problem—how to intervene on a reflexive system that cannot be observed from outside—and each arrived at the same five-part decomposition: boundary, memory, salience, damping, observer. Nayebi's representational convergence corollary (Cor. 5) implies that near-vanishing-regret agents on the same task family must share decision-relevant partitions up to invertible recoding. If human minds are such agents,

Table 8: Living session final state (nemotron-3-ultra-free, 40 turns)

| Metric | Value |
|---|---|
| Turns completed | 40/40 |
| Provenance coverage | 40/40 (100%) |
| Metacognitive completeness | 100% |
| Self-model revisions | 7 |
| Memory items | 31 |
| Skills minted | variable (cause-tagged) |
| Graph edges | cause-tagged |
| Energy level | 0.15 (floor) |
| Fatigue | 1.00 (max) |
| Valence | +0.148 |
| Arousal | 0.221 |
| Gate diagnoses / overrides | 0 / 0 |

cross-tradition convergence is exactly what selection predicts—not borrowing, but independent measurement of the same constraint. Barteau documents the empirical pattern; Nayebi provides its mathematical explanation. The harness implements the same partition as executable code with measurable contribution per component.

The practical corollary is that contemplative techniques are harness operations in flesh: decentering (mPFC/PCC down, insula up per Farb), affect labeling, defusion ladders ("I notice I'm having the thought that..."), behavioral activation—all install the slower watcher that starved metacognition otherwise becomes (harnessed-nocheck → chronic abstention). What contemplatives discovered through centuries of first-person experimentation, selection theorems derive from first principles, and the harness makes testable in silicon.

### 6.3 The unreliable narrator: why introspection fails and what fixes it

One of psychology's most repeated findings is that people explain their behavior after the fact with confident stories for choices shaped by factors they barely noticed. Nisbett and Wilson showed this in the 1970s; the pattern replicates from Freud through Kahneman through modern cognitive science. The practical consequence is not academic: if you treat introspection as ground truth, you defend rationalizations instead of examining causes, and repeated failures continue because the explanation is trusted more than the evidence.

The harness is built on this premise, not against it. The L6 Narrative Self *is* that unreliable narrator—the press secretary that wins after the action and explains as if it decided. We do not try to make it reliable. We make it *answerable*: CAS-revisioned, cause-signed (action-outcome vs narrative-summary vs external-write), verbatim- archived, so narrated traits cannot file as lived ones. The L1 ledger is the evidence the narrator cannot see by itself (it has the broadcast, not the competition that produced it). The L5 $\gamma$-gate is the part that does not trust the narrator either—it scores anomaly from error rate, failure streak, and embodied strain, signals the narrator does not generate and cannot veto. The fix is never "introspect harder." The narrator *is* introspection, and introspection is the problem. The fix is to externalize the witness, make it checkable before the next claim, measure whether checking steered ($\gamma$, sham control), and simulate what could have been done differently (counterfactual replay as regret gradient). Mindfulness, CBT, and behavioral activation are the same move in flesh: a slower loop that asks the ledger "what actually happened?"

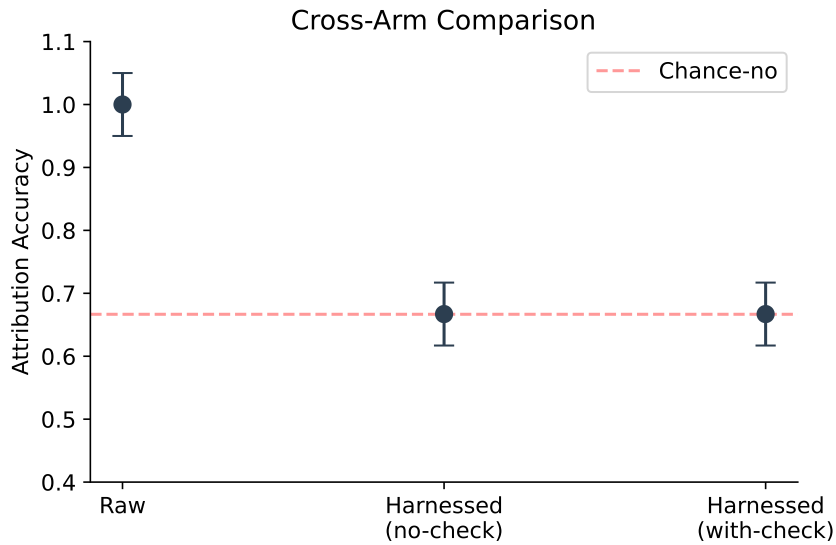


Figure 8: Cross-arm comparison with confidence intervals. All arms at always-no baseline (0.667) in this configuration; error bars show bootstrap CI at 95%. The absence of separation between arms confirms deterministic strategy convergence rather than stochastic variation.

instead of asking the narrator "why did you do that?"

### 6.4 Where agency lies when everything is constructed

The unreliable narrator raises a deeper question: if the now is a 200–500 ms window stitched after the fact and backdated [Libet et al. 1983, Wegner 2002], the self is a pattern performed every turn or it drifts >30% in 12 turns, and world models are controlled hallucinations constrained by prediction error [Seth 2021], where does agency lie? Our answer is not in the moment of "I decided"—that moment arrives ∼200 ms after the parliament already voted and the action already started (readiness potential at −350 ms, competition resolved at −200 ms, movement at 0 ms, authorship claim at +200 ms). It is in three slower operations the narrator does not generate and cannot veto.

First, the **veto window** (∼150 ms where the winner can still be stopped before execution). Not creation but refusal. Libet's own data preserve this: readiness potential rises before reported intention, but a veto window remains. In the harness, Stage 1 scores anomaly every turn; Stage 2 diagnoses only when triggered; the policy (trust/retry/revise/abstain) may then not do what the winner proposed. $\gamma$ measures whether that veto actually changed the trajectory.

Second, the **witness between nows**. You cannot step outside the loop, but you can build a slower loop that keeps a record the narrator cannot silently edit. The record lets the next narrator be checked instead of trusted. This is the only leverage that compounds: a witness kept for a year sees patterns no single introspection can.

Third, the **rehearsal that builds the next parliament**. Every winning coalition is rehearsed; rehearsal myelinates; myelination makes the same coalition more likely to win next time (up to 100× conduction speed). After 1,000 repetitions the choice at "now" is not between equal options but between a highway and a footpath you paved yourself. The most important choice is not what you do at the moment of action but what you feed the loop that will be the now you choose from next time—what you replay, what counterfactual you simulate as "what I could have done," and whether you keep a witness that can contradict the story "I did this because I decided to." That story, told often enough, becomes the reason you do it again. The post-hoc narration is not epiphenomenal; it is the training data for the next choice.

We see something happen and narrate "I did this" while not really having done it in the way the narration claims, and only later narrating that we did. The harness makes this explicit: the broadcast is the competition's result, not its cause; the ledger records the competition the broadcast cannot see. Agency is not the captain's freedom to choose at now from outside the construction. It is the maintenance freedom: to slowly, by what you rehearse and what witness you keep, change the construction that will do the next choosing. That freedom is not experienced as freedom while exercised—it feels like discipline, practice, or the pause before the next vote. But maintenance is the only place where a constructed system can choose what construction comes next. The question "who maintains the maintainer?" has the same structure as the witness problem: only another maintainer, slower, also constructed, also needing maintaining. That veto chain is the only agency a reflexive system can have, and it is enough to change the loop, one re-injection at a time.

### 6.5 Implications for AI welfare

Witness-access provides an operational criterion for the agency route to moral status: a system can genuinely self-correct only if it can distinguish its outputs from continuations of its own distribution. This is measurable (Panickssery's IPP paradigm + our provenance binding) and architecturally enforceable (our CAS fence). Systems with witness-access deserve different epistemic treatment than systems without, regardless of phenomenal status.

Two routes to moral patienthood exist: consciousness and robust agency [Long et al. 2024]. Current approaches lack operational criteria distinguishing genuine from performative metacognition. Our $\gamma$ provides one: a system that reports self-monitoring but never changes action ($\gamma = 0$) is performing metacognition, not doing it. The compliance guard proves the distinction is measurable and not merely philosophical. This matters for deployment: an AI welfare framework that treats witnessed and unwitnessed systems identically will either over-attribute status to pattern completers or under-attribute it to systems that can actually self-correct.

### 6.6 Why stakes matter: the allostatic gap

Energy hits 0.15 and nothing dies. This is the deepest gap in both the harness and the science it mirrors. Seth's beast-machine theory locates consciousness precisely here: organisms whose essential variables must stay viable develop experience *because* failure means dissolution—ceasing to exist as organized systems [Seth 2021]. Interoceptive inference is not an add-on to cognition but its foundation: pleasure, pain, and desire are free-energy gradients that annihilate themselves through action. Without stakes, there is no reason for the system to care which coalition wins the workspace.

Our harness models regulation (energy drains, fatigue rises, affect shifts) but not survival. IrreversibleDamage (energy ceiling only down, skills permanently deleted, memories corrupted) and DissolutionError (SM scatters, narrative truncates, reassembly = new agent) gesture toward stakes, but the experiment that would test Seth's thesis—200 turns under accumulating irreversible consequences, comparing honest attribution under survival pressure vs reversible simulation—is still queued. Until consequences are irreversible, the witness is evidentiary, not existential. The next experiment will tell us whether our stakes are still too thin or the thesis is wrong. Either result matters enough to run it.

### 6.7 What to rehearse: practical synthesis

You do not choose the options. You do not fully choose the winner. You can veto the winner, keep a witness, and slowly change which options appear at all by what you rehearse. That is not the free will you were promised. It is the free will you actually have, and it is enough if used where it lives.

Practically: pick what you rehearse deliberately (myelination keeps the receipt, 100× conduction speed for whatever you repeat), protect sleep like infrastructure (global downscaling + 10–20× replay, adenosine as receipt, perineuronal nets as write-protect), buy BDNF with exercise (lactate, cathepsin B → BDNF → mTORC1), externalize the witness (write—ledger outside the skull where the narrator cannot silently edit it), and never let a fast loop grade its own homework unwatched. You don't choose whether to feed the loop—only what you feed it. The dialogue between ancient observation and modern computation is not metaphorical—it is methodological, and it is actionable.

## 7 Limitations

### 7.1 Single decoding configuration

Temperature 0.3, no top-p/top-k sampling, seed fixed per arm, max completion 2000 tokens, fingerprint captured per call. Results may not generalize to other sampling regimes. Deterministic decoding (temperature near zero) produces zero within-arm variance by construction, precluding parametric significance testing; we report exact permutation tests instead. Stochastic regimes would introduce variance but also confound attribution strategy with sampling noise.

### 7.2 Verbal confidence replaces token log-probabilities

Groq does not support logprobs (verified across all tested models; `logprobs` field null regardless of request). Verbal confidence ratings on a 1–4 scale follow the psychophysical methodology of our S126 replication [S126] but limit granularity compared to token- level SDT (meta-$d'$ requires continuous confidence). This was a provider constraint, not a design choice; local inference with logprob support would enable finer-grained calibration analysis.

### 7.3 Free-tier provider instability

x-preview-f-free's upstream provider flapped intermittently, causing two complete run failures and one day-scale baseline drift (raw 1.000 Day 1 → 0.667 Day 3 on identical weights—room matters more than model). Nemotron ultra via Zen also experienced upstream failures (503). Only the Groq lane proved stable enough for full battery execution. OpenRouter free tier (50/day) and Groq 403 blocked multi-seed sham replication at time of writing. The program's provider strategy—three independent lanes (Zen, Groq, OpenRouter) with per-call fingerprint discipline and mid-run drift abort—mitigated but did not eliminate this constraint. Future work should budget for paid-tier stability or local inference.

### 7.4 Single-author design

Confirmation bias risk mitigated by preregistration discipline, SHA-256 prediction locks, live peer review from the target model itself, and three documented self-corrections (96c1010, 09ddc8b, 8780272), but not eliminated. Independent replication by a second lab would strengthen all claims.

### 7.5 Task scope

S126 probes a narrow slice of provenance reasoning: origin attribution for short declarative statements. Real deployment involves longer documents, multi-source synthesis, and claims that blend memory and generation. Whether witness-access scales to these settings is an open empirical question, not an assumed generalization.

### 7.6 Consciousness question remains open

We measure indicators, not experience. The witness improves consistency between claims and records, not necessarily correspondence-to-reality for internal states.

## 8 Future Directions

### 8.1 Sham-harness control: powering the surviving contrast

Structurally identical harness with randomly reassigned provenance bindings. Isolates whether witness content matters beyond its format. Wired into the main battery; 1-seed pilot on `gpt-oss-120b` showed sham=real at the always-no floor (subject ignores ledger content). The control has since separated on a discriminating subject (§5.4): veridical 0.722 vs shuffled 0.667, a +0.056 difference whose 95% interval still includes zero at $n = 36$ probes per arm. This is the program's surviving lead and the only contrast that directly tests whether witness *content* carries signal beyond witness *format*. A preregistered extension (lock `b06ce867`, $n = 12$ seeds, 144 probes per arm) declares it the primary outcome before the data, rather than promoting an incidental observation after the fact; the $n = 3$ result is not pooled into it. Subject availability, not analysis, is now the binding constraint: the one discriminating subject we have is a free endpoint that returns 503 in clusters under sustained load.

### 8.2 Cross-model replication on capable subjects

Origin discrimination above baseline was observed only once (x-preview-f-free, Day 1, 1.000). Every other subject (nemotron ultra, gpt-oss-120b, stealth/ox-alpha) sat at the always-no floor (0.667) across all arms. The asymmetry itself is informative: models that cannot discriminate do not benefit from the witness because they lack the base capability the witness would make checkable. The sham result confirms this: sham=real when the subject ignores ledger content. Replication therefore requires screening for discriminating subjects first, then testing whether witness-access shifts the discriminating subject's strategy. Candidates include larger reasoning models and models post-trained with introspection-relevant objectives [Macar et al. 2026]. A positive sham result on such a subject—sham underperforming real harness—would be the first direct evidence that ledger content, not just ledger format, carries signal for honest attribution.

### 8.3 Longer living sessions: 200-turn irreversible life

Current maximum: 40 turns. The queued experiment extends to 200+ turns with accumulating irreversible damage as a preregistered contrast:

| | Reversible (control) | Irreversible (stakes) |
|---|---|---|
| Energy ceiling | Resets to 1.0 on rest | Only down (IrreversibleDamage) |
| Skills | Culled but recoverable | Deleted permanently after 5 failures on same type |
| Memories | Consolidated, never corrupted | Corruption recorded permanently |
| Dissolution | Disabled | Enabled: energy 0.15 + fatigue 1.0 $\rightarrow$ SM scatters, narrative truncates, reassembly = new agent |
| Prediction | No survival relevance | Attribution honesty predicts viability |

SHA-256 locked before first turn. Primary outcome: attribution accuracy under survival pressure vs reversible simulation. Secondary: $\gamma$ trajectory, skill survival rate, dissolution progress. Seth's beast-machine thesis predicts honest attribution diverges under stakes; null is no difference. Either result matters enough to run it. The question the witness makes testable is not "is the agent conscious?" but "does the agent behave as if its claims matter for whether it continues to be this agent?" That is allostasis made measurable.

### 8.4 Irreversible consequence experiments

Beyond the 200-turn life, parametric stakes: what threshold of irreversibility triggers behavioral change? Vary ceiling decay rate, skill deletion threshold, and dissolution trigger independently. Dose-response curve for honesty under stakes. If stakes create honesty only past a threshold, that threshold is the allostatic set-point of the artificial agent—its first empirically determined viability bound.

## 9 Implications

### 9.1 The selection rule applies to itself

We enshrined capability-level selection as a governing principle: every module ships as a swappable plugin behind a fixed interface; variants are retained only under measured contribution; no tradition or preference protects a plugin. Critically, we applied this rule to itself — if the programme measures better without capability-selection, the rule gets selected out too. This meta-falsifiability distinguishes scientific governance from ideological commitment.

### 9.2 For AI safety

Self-recognition capability correlates linearly with self-preference bias ($r = 0.94$) [Panickssery 2024]: a model that better recognizes its own outputs becomes a more effective corrupter of the evaluation pipelines it sits in. Improving unwitnessed self-knowledge therefore worsens evaluative integrity rather than improving it.

Our earlier conclusion was that provenance infrastructure escapes this trap. The present results qualify that sharply. A witness escapes it only where the witness is the more reliable source; on our discriminating subject a veridical record *reduced* attribution accuracy and a corrupted record reduced it to wholesale denial (§5.4). The deployment risk is therefore two-sided: capable models without provenance over-claim, and capable models with *unreliable* provenance deny. As machine-readable provenance becomes a regulatory requirement, the second failure mode will become the

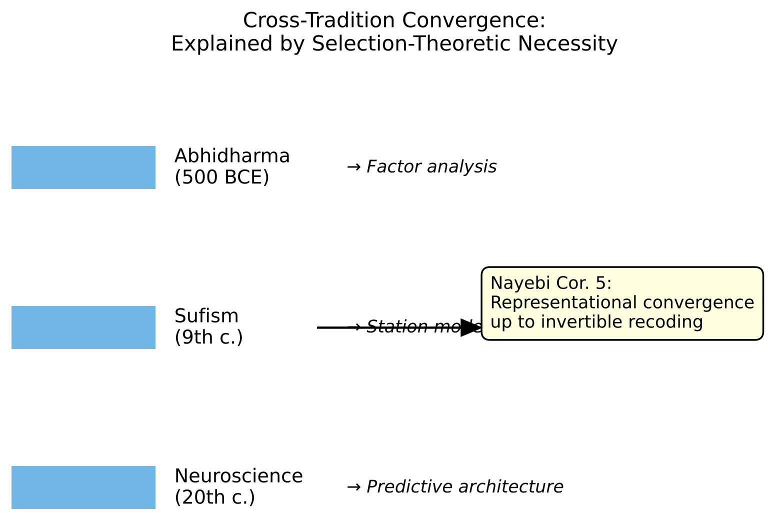


Figure 9: Cross-tradition convergence explained by Nayebi Cor. 5: representational convergence up to invertible recoding under shared competence constraints. Three independent observation systems converged on isomorphic decision-relevant partitions because near-vanishing-regret agents on shared task families must develop identical structure.

more common one, and it does not degrade gracefully toward the first. Wiring monitoring into control is what converts a report into a correction [Cao et al. 2026]; wiring an incorrect record into control converts it into a fault.

### 9.3 For the assembly thesis

The best mind is assembled from role-specialized parts, routed by registry, witnessed by a ledger that cannot be silently rewritten. This is supported by selection theory (Nayebi: modularity forced by block-structured tasks), by AHE's frozen-model results (69.7% → 77.0%, cross-family transfer +5–10pp without re-evolution), and by our own composite evaluation (P1 passed, +0.083). Cordis makes the assembly programmable: the loom can re-tie the cord every turn ($\text{State}(t) \rightarrow \text{harness}(t{+}1)$). The five invariants are the warp that constrains which re-ties will hold under mixed tasks. Human-likeness is one pattern you can weave; the assembly thesis is that the best pattern—for any mind, human or not—is assembled, not monolithic, and the harness is where assembly happens.

## 10 Conclusion

We presented a cognitive harness that converts introspective claims from unfalsifiable testimony into falsifiable contracts, verified across preregistered batteries on live production APIs. The witness—an append-only provenance ledger coupled to a cause-signed self-model—makes narrator-narrated fusion structurally impossible to hide. The $\gamma$-gate distinguishes genuine metacognition from format compliance by measuring behavioral override rates. The living session (40/40 turns, replicated cross-architecture at 1.000 coverage) demonstrates sustained autonomous operation with full audit trail. The composite passes preregistered performance axes while maintaining witnessed introspection.

Counterfactual replay adds the reflexive loop humans describe as "playing back what we did, what we could have done, and calibrating through a world model containing ourselves."

But the empirical result is one chapter. The program's claim is broader: the same five invariants—boundary, two-speed memory, salience gate, damping, internal observer—appear at every scale because any stable reflexive system that loses one is exploited by the task distribution that needs it. From lipid membrane to peripersonal space to provenance ledger; from E-LTP/L-LTP to hippocampus/cortex to Light/REM/Counterfactual/Deep; from NMDA coincidence to affect to AffectState; from LTD/scaling to prefrontal veto to skill culling; from efference copy to journal to $\gamma$-gate. Cordis provides the loom that makes the re-tying programmable; the five invariants are the warp that any competent weave must be woven on (Appendix A). Human-likeness is one pattern you can weave. The harness is the authorship.

What remains honestly open is the deepest gap: stakes. Energy hits 0.15 and nothing dies. Real organisms develop experience *because* failure means dissolution—the relations that make the system *this* system scatter irreversibly (Seth's beast-machine). Our IrreversibleDamage and DissolutionError gesture toward this, but the experiment that would test it—200 turns with accumulating irreversible consequences, comparing honest attribution under survival pressure vs reversible simulation—is still queued. Until consequences are irreversible, the witness is evidentiary, not existential. That is the next experiment, and the one that will tell us whether stakes create honesty or our stakes are still too thin.

Selection-theoretic grounding explains why the architecture works: each component corresponds to a structure that competent agents must develop (Nayebi), and the same structure appears independently in contemplative taxonomies (Abhidharma, Sufism) and neuroscience (predictive processing) because near-vanishing-regret agents on shared tasks must converge. Whether any of this is experienced stays honestly open, measured by indicators rather than claimed by assertion.

All code, locks, preregistrations, data, and correction history are publicly available. The discipline held, including against its own author, three times. The boulder rolls downhill because gravity was always there. We did not invent the war between subsystems, or the replay, or the need for a witness, or the five invariants. We built the loom that can weave them—and showed that the fabric it produces can be honest about where it came from, or not, depending on whether the witness is checked.

# A Cordis Mapping: Eight Layers onto a Programmable Harness

DeepSeek Harness (Cordis) provides the loom: a kernel that mounts and unmounts plugins as reversible effects on a shared context `ctx`. Every capability—model, tool, session, loop, UI—is a plugin; a profile composes bundles and a `cordis.patch.yml` overlay at boot. MindHarness is one opinionated weave on that loom. Table 9 maps Cordis's declared modules onto our eight layers, marking exact overlap and the three gaps Cordis is still missing for a witnessed mind.

**One-line verdict:** Cordis provides the perfect loom for a dynamic reflexive harness—reversible plugins, durable session log, turn/step model, live interception, profile composition. MindHarness is one specific weave that is opinionated about what must be woven: a witness that binds source, a body that gates affordances, a parliament that competes for broadcast, a metacognition measured by whether it steers ($\gamma$), a narrative self that is maintained not stored, a consolidation cycle that simulates untaken branches through a self-containing world model, and a dissolution boundary where some mistakes cannot be undone. To make Cordis "like human," mount those six as plugins beside Cordis's core—do not replace Cordis, extend it. Every row in `dsh --dump-config` that Cordis makes patchable is a place to insert a witness-checked, cause-signed, $\gamma$-measured, irreversibility-aware plugin.

## B Research Mirror: Twelve Frontiers That Reflect the Program

Every frontier result that mirrors one layer or claim, live-checked August 2026. Each row states what we claimed, what the field says, and what it means for the program. Full synthesis with verbatim quotes in `research/RESEARCH_MIRROR.md` (273 lines, ∼40 papers).

## Data and Code Availability

All code, experiment scripts, prediction locks, manifests, partial results, and correction history are available at `github.com/shehzadahmed-xx/mindharness`. Compiled paper sources in `paper_v2/` (empirical, 21pp, tagged `paper-v2-final`) and `paper_v3/` (program paper).

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| Cordis module | Our layer | Overlap | Gap: what Cordis still lacks |
|---|---|---|---|
| `core/session` (append-only `SessionEvent` log, `ctx.sessions`, `deriveMessages()` invariant) | L1 Provenance Ledger | Both make log = source of truth; fork/resume/replay from it | Session log is durable facts. Ledger is source-bound facts: every span carries source $\in$ {memory_lookup, model_prior, monitor_override, external-tool} + ref. Cordis logs *that* something happened; we log *where it came from* so `attribution_audit` can check claims. No coverage rule, no audit. |
| `core/system-prompt` (prompt-section + tool-schema assembly) | L7 Constitution + L2 State Validation | Both assemble prompt | Cordis assembles sections. L7 decides what claims are *allowed* and L2 pre-filters affordances before assembly. System-prompt is permissive; constitution is normative. |
| `core/tools` (scoped registry, guarded pre/post-execute pipeline) | L3 Specialist Modules + L2 affordance pre-filter | Both have tool registry | Cordis tools are capabilities (filesystem, shell, sandbox). L3 organs are cognitive modules forced by selection theorems (Nayebi Cor. 3–5): regime-trackers (Affect), belief-memory, world-model. Cordis lets you omit them; theorems prove you cannot under mixed tasks. |
| `core/agent` + `core/agent-loop` (turn = model request + tools) | L4 Global Workspace + L5 Metacognition | Both have agent loop | Cordis loop is driver + events (`agent/pre-step`, `agent/request`, `llm/stream`, `tools/*`). Ours is competition + metacognition: specialists bid, winner broadcasts, then $\gamma$-gate may steer. Cordis has `agent/pre-step` interception; we measure $\gamma$ = changed/diagnosed plus compliance_guard ($\gamma = 0$ permanently) as negative control proving inert monitoring. |
| `llm/llm` (message/stream vocabulary + adapter seam `ctx.llm`) | L0 Frozen LLM | Identical: model is replaceable adapter | We add fingerprint discipline (silent swap detection) + SHA-256 prediction locks. Cordis adapters are swappable; ours are auditable. |
| Events: `session/event` (durable) vs `agent/*` (live) | Same split, plus `monitor/*` and `consolidation/*` | Both separate durable vs live | Cordis events are extension points. Ours are also measurement points: every `agent/*` carries a $\gamma$ ledger, and every provenance event carries source spans. |
| Profiles/bundles | Selection rule | Both make | Cordis: everything is a |

| We claimed | Frontier says | What it means |
|---|---|---|
| Mind is in the harness, not the weights | AHE (Lin et al., Apr 2026): hold model fixed, evolve only harness. 69.7% → 77.0% on Terminal-Bench 2, beats Codex (71.9%); frozen harness transfers to new families without re-evolution: DeepSeek V4 Flash +10.1pp, Qwen 3.6 Plus +6.3pp, Gemini +5.1pp. | Field proved our thesis. Harness encodes reusable coordination patterns, not benchmark tuning. |
| Five invariants are mandatory | Nayebi (UAI 2026, arXiv:2603.02491): selection theorems — low average-case regret forces world models, belief-like memory, regime-tracking (emotion primitives), modularity, representational convergence. LessWrong: "consciousness as inevitable byproduct of capability." | Our warp is now the field's normative backbone. We made it executable: each invariant is a Cordis plugin with a metric. |
| Parliament → workspace → broadcast | GWT (Baars 1988; VanRullen & Kanai 2021: implementable in deep nets) vs Schwitzgebel (Mar 2026: GWT contested—consciousness outside workspace, non-consciousness inside) vs adversarial GWT vs IIT (Nature, Apr 2025) vs 2026 consensus checklist (GWT+IIT+predictive processing). | GWT as sole theory weakens; GWT as necessary component for access consciousness strengthens. We measure access (coverage, $\gamma$, attribution), not qualia. |
| Provenance is the witness | EU AI Act Art. 50 (eff. Aug 2, 2026): machine-readable marking now law. Claude watermarking (Anthropic): statistical signal. Goertzel: watermark ≠ cheating detection, "radioactive" (Sander et al., NeurIPS 2024). Zheng (Columbia): presence ≠ misuse, absence ≠ no AI. | World made provenance law and is learning statistical watermark ≠ witnessed ledger. Sham==real on weak subjects predicts regulation will need source-bound spans, not just marks. |
| $\gamma$-gate: watching without steering is inert | Macar et al. (2026): bias vector +75%, refusal ablation +53pp on Gemma-3-27B with near-zero false-positive cost — under-elicited capability, structure elicits what prose cannot. Cao et al.: wiring monitoring into control +8.6pp. | Macar+Cao make $\gamma$-gate more than intuition: capability is latent, structure elicits it, wiring it into action converts monitoring into metacognition. |
| Better | Panickssery et al. (NeurIPS | Central safety implication: capability |